AdifySkills

n8n Automation Mastery
+91 83838 91716

CLASS NOTES

🏗️ Module 7 · Real-World Project Sprints

DAY 23 of 25

Project Sprint 3: Finance & Operations Autopilot

"Back-office jo khud chalta hai."

⚡ AAJ HUM YEH BANAYENGE

Finance Autopilot: Invoice Approval + Monthly P&L

Do flows ek workflow mein: (1) vendor invoice → AI extraction → ₹10,000 se kam auto-approve, upar wala manager ko email par Approve/Reject → Payables sheet. (2) Har mahine ki 1 tareekh ko sheet se P&L summary email.

Gmail | AI Extractor | Sheets | Telegram

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Back-office (finance, HR, ops) mein sabse zyada repetitive kaam — aur sabse kam competition.

🎯 Aaj ke baad aap yeh kar paoge

01 Invoice intake & AI extraction
🌍 Daakiya bill laaya, padh ke diary mein.
⚙️ Gmail Trigger (has:attachment invoice) → AI extract → Sheet.

02 Approval chains & payment reminders
🌍 ₹10k tak bhaiya, upar papa.
⚙️ Switch amount → manager / CFO approval (Slack buttons).

03 Bank/payment reconciliation basics
🌍 Doodh wale ka hisaab milana.
⚙️ Bank statement rows vs invoices → match by amount+date → mismatch list.

04 HR onboarding automation
🌍 Naye bachche ka school mein pehla din.
⚙️ Form submit → Google account, Slack invite, welcome email, task list.

05 Management dashboards
🌍 Mahine ka report card.
⚙️ Schedule month-end → aggregate → chart → email PDF.

06 Client handover & maintenance SLA
🌍 Gaadi ki servicing contract.
⚙️ SLA: response time, monthly fee, kya cover hai, kya nahi.

⏱ Class ka plan (2 ghante)

15m	75m	20m	10m

`0:00-0:15` **Brief & Architecture**
🪄 *Mummy ki diary mein har kharcha likha hota hai aur mahine ke end mein total. Robot yahi diary apne aap bharta hai.*

`0:15-1:30` **Build Sprint**
🪄 *Marathon ek saath nahi daudte — har kilometer par paani peete hain. Har phase ek kilometer hai.*

`1:30-1:50` **Demo & Review**
🪄 *Science fair — model dikhao, batao kaise chalta hai, sawaalon ka jawab do.*

`1:50-2:00` **Capstone brief**
Capstone topics finalize
🪄 *Project ka agla page — ghar par likhna hai, kal sabko dikhana hai.*

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Approval chain

Amount ke hisaab se manager → finance → CEO.

🪄 **Analogy:** Leave application flow

CONCEPT 02

Reminders

Due date se pehle/baad automatic nudges.

🪄 **Analogy:** Kisi ka udhaar yaad dilana

CONCEPT 03

Reconciliation

Bank/payment entries ko invoices se match.

🪄 **Analogy:** Hisaab milana

CONCEPT 04

SLA & maintenance

Delivery ke baad support ka agreement.

🪄 **Analogy:** AMC (annual maintenance)

🔑 Aaj ka main idea

Finance/Ops autopilot = invoices, approvals, reminders, reconciliation aur reports ko jodna. Rule-based steps + AI extraction + human approval jahan paisa jaaye.

- Vendor invoice → manager approval → payment reminder
- Naye employee ka onboarding: email, Slack, docs
- Monthly P&L summary CEO ko

📖 Ek kahani se samjho: Ghar ka hisaab-kitaab robot

1. 📨 **Bill aaye email mein** = Invoice intake
2. 🔍 **Robot ne padha** = AI extraction
3. 👨 **Papa ne approve kiya** = Approval chain
4. ⏰ **Due date se pehle yaad** = Payment reminders
5. 📊 **Mahine ka report card** = P&L dashboard

🪄 Ek line mein: "Mummy ki diary mein har kharcha likha hota hai aur mahine ke end mein total. Robot yahi diary apne aap bharta hai."

🗺️ Aaj ka workflow: Finance Autopilot: Invoice Approval + Monthly P&L

Data left se right chalta hai. Har dabba ek n8n node hai.



💰 Isse kitna kama sakte ho?

SMEs, startups (finance autopilot) — aam taur par **₹40,000 – ₹1,00,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1. ⚡ **Gmail Trigger**
   Vendor invoice emails

2 ⚡ **Information Extractor**
Invoice fields

3 🔀 **Switch**
Amount < 10k auto, 10k–1L manager, > 1L CFO

4 💼 **Slack**
Approval request

5 ⚡ **Schedule Trigger**
Daily due-date reminders

6 📊 **Google Sheets**
Monthly P&L summary + email

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- [ ] Connect: **Gmail (Google OAuth)**
- [ ] Connect: **OpenAI API key**
- [ ] Connect: **Google Sheets (Google OAuth)**
- [ ] Sheet tabs: "Payables" (vendor, invoice_no, total, due_date, status) aur "Ledger" (date, type, category, amount)
- [ ] Manager / owner emails badlo

✅ Test kaise karein

1. ₹5,000 aur ₹25,000 ke do sample invoices email karo
2. Bade wale par approval email aana chahiye
3. P&L flow ko manually execute karke dekho

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😫 Pehle (manual)
Accountant ka roz ka data entry + follow-up — **240 min**
240 min

⚡ Ab (n8n)
Exceptions check karna bas — **30 min**
30 min

⏱ Accountant ka ~85% time analysis par, entry par nahi.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo — 📋 Select karke copy karo

```
{{ $json.total < 10000 ? "auto" : $json.total <= 100000 ? "manager" : "cfo" }}
```

📦 Sample data — input se output tak

📥 Input 📋 Select karke copy karo

```
{
  "vendor": "Cloud Hosting Co",
  "total": 45000,
  "due": "2026-12-10"
}
```

📤 Output 📋 Select karke copy karo

```
{
  "route": "manager",
  "approved": true,
  "reminder": "2026-12-08"
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Do flows ek workflow mein: (1) vendor invoice → AI extraction → ₹10,000 se kam auto-approve, upar wala manager ko email par Approve/Reject → Payables sheet. (2) Har mahine ki 1 tareekh ko sheet se P&L summary email.

Flow: Gmail Trigger (Vendor invoice) → PDF + AI extract (Amount, vendor) → IF (₹10k se upar?) → Gmail send & wait (Manager approval) → Sheets + Monthly P&L (Payables + report)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt 📋 Select karke copy karo

```
I'm building the n8n workflow "Finance Autopilot: Invoice Approval + Monthly P&L" (nodes: Vendor Invoice, PDF Text, Extract, Extract Model, Above 10k?, Manager Approval, Set Status, Payables, 1st of Month, Read Ledger, P&L Summary, Email P&L).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ Workflow JSON
12 nodes · ready to import
adifyskills.com/workflows/day-23-finance-ops-autopilot.json

🌐 Lesson page
Copy buttons, quiz aur video links
adifyskills.com/day/23

Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Gmail (Google OAuth)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "OpenAI API key" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Google Sheets (Google OAuth)" connect karo.
Time par apne aap nahi chala?	"Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo.
AI node par 401 / 429 error?	401 = API key galat. 429 = credit khatam ya bahut jaldi requests. OpenAI billing page check karo.
"Sheet not found" / column khaali?	Sheet ID aur tab ka naam (capital/small bhi) check karo. Pehli row mein headers hone chahiye.

⚠️ Yeh galtiyan sabse common hain

- ! Approval history save na karna (audit trail)
- ! Timezone galat — reminder raat 3 baje
- ! Client handover mein credentials ownership clear na karna

⭐ Pro tips — time bachane wali tricks

- ★ Har approval ka who/when log
- ★ Client ke apne accounts par credentials — aapke nahi
- ★ Maintenance retainer proposal ready rakho

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Ops autopilot: invoices → extraction → approval → reminders → monthly P&L summary.

📝 Homework

Capstone project proposal submit karo (problem, solution, architecture diagram).

- ☐ Homework ho gaya

BEGINNER
Due-date reminder

INTERMEDIATE
Approval chain

ADVANCED
Payment reconciliation with fuzzy matching

🧩 Aaj ka challenge — pehle khud try karo

🎯 Due date se 3 din pehle vendor payment ka reminder finance team ko jaaye.

💡 Hint 1
Roz chalne wala ek alag flow chahiye jo Payables padhe.

💡 Hint 2
Schedule daily → Sheets read Payables → Filter due_date = aaj + 3 din → Gmail.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Amount ke hisaab se 3 raaste?
- (A) Limit
- (B) Switch
- (C) Merge
- (D) Sort

Q2. Daily reminders ka trigger?
- (A) Form
- (B) Chat
- (C) Schedule Trigger
- (D) Webhook

Q3. Audit ke liye kya save karein?
- (A) Kuch nahi
- (B) Sirf total
- (C) Emoji
- (D) Approval history

Q4. Delivery ke baad support agreement?
- (A) SLA / maintenance
- (B) SRS
- (C) API
- (D) SQL

📖 Aaj ke naye shabd

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

Switch
Kai raaston mein se ek chunna — jaise traffic police kisi gaadi ko alag lane mein bhejta hai.

Schedule
Fixed time par workflow chalana — roz, har ghante, har Monday.

✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 24

Freelancing, Agency & Client Delivery

"Skill ko income mein badlo."

☐ **Day 23 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → B
Switch

Q2 → C
Schedule Trigger

Q3 → D
Approval history

Q4 → A
SLA / maintenance

✅ Challenge ka solution

Schedule (daily 10am) → Read Payables → Filter {{ $json.due_date }} equals {{ $now.plus({ days: 3 }).toFormat('yyyy-MM-dd') }} → Gmail reminder.

📚 Official docs

- Finance templates ↗
- Extract from File ↗
- Schedule Trigger ↗
- Switch node ↗
- Information Extractor ↗

▶ Videos

- n8n invoice approval workflow ↗
- n8n finance automation ↗

Doubt poochho
WhatsApp: +91 83838 91716
wa.me/918383891716

Day 23 lesson page
Quiz, copy buttons, videos
adifyskills.com/day/23

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